Practical No. 4

Q.a

Aim:

Write a python code to create Numpy Array.

Code:

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50])
print("NumPy Array:")
print(arr)
```

Output:

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/Deepak Sahani/OS/practical_4_Q.1.py ===============
Deepak Sahani
NumPy Array:
[10 20 30 40 50]
>>>
```

Q.b

Aim:

Write a python code to demonstrate basic operations on single array.

Code:

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50])
print("Deepak Sahani")
print("Original Array:", arr)
print("After Adding 5:", arr + 5)
print("After Subtracting 5:", arr - 5)
print("After Multiplying by 2:", arr * 2)
print("After Dividing by 2:", arr / 2)
print("Square of Array:", arr ** 2)
print("Sum:", np.sum(arr))
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))
print("Average:", np.mean(arr))
```

Output:

```
*IDLE Shell 3.14.6*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/Deepak Sahani/OS/Practical_4_Q.2.py ===============
Deepak Sahani
Original Array: [10 20 30 40 50]
After Adding 5: [15 25 35 45 55]
After Subtracting 5: [ 5 15 25 35 45]
After Multiplying by 2: [ 20  40  60  80 100]
After Dividing by 2: [ 5. 10. 15. 20. 25.]
Square of Array: [ 100  400  900 1600 2500]
Sum: 150
Maximum: 50
Minimum: 10
Average: 30.0
```

Q.c

Aim:

Write a python code to create array with 10 elements and slice element from 1st to 5th element.

Code:

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
print("Deepak Sahani")
print("Original Array:")
print(arr)
slice_arr = arr[0:5]
print("Sliced Array (1st to 5th element):")
print(slice_arr)
```

Code:

```
IDLE Shell 3.14.6
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/Deepak Sahani/OS/Practical_4_Q.3.py ===============
Deepak Sahani
Original Array:
[ 1  2  3  4  5  6  7  8  9 10]
Sliced Array (1st to 5th element):
[1 2 3 4 5]
```

Q.d

Aim:

Write a python code to sort an array alphabetically.

Code:

```
import numpy as np
arr = np.array(["Banana", "Apple", "Orange", "Mango", "Grapes"])
print("Deepak Sahani")
print("Original Array:")
print(arr)
sorted_arr = np.sort(arr)
print("Sorted Array:")
print(sorted_arr)
```

Name:Deepak Sahani
Roll no.:S108

Output:

```
IDLE Shell 3.14.6
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/Deepak Sahani/OS/Practical_4_Q,4.py ===============
Deepak Sahani
Original Array:
['Banana' 'Apple' 'Orange' 'Mango' 'Grapes']
Sorted Array:
['Apple' 'Banana' 'Grapes' 'Mango' 'Orange']
```

Q.e

Aim:

Write a python code to create a filter array that will return maximum values from an array.

Code:

```
import numpy as np
print("Deepak Sahani")
arr = np.array([10, 25, 15, 40, 30, 40, 20])
max_value = np.max(arr)
print("Original Array:", arr)
print("Maximum Value:", max_value)
```

Output:

```
IDLE Shell 3.14.6
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/Deepak Sahani/OS/Practical_4_Q.5.py ===============
Deepak Sahani
Original Array: [10 25 15 40 30 40 20]
Maximum Value: 40
```